rotation cycles. Sequential reads and writes can further be augmented by applying readahead or prefetch algorithms to predict or preempt the next read or write, and hence speed up the IO cycle further. In case of random IOs, the time to seek a block increases the time to perform read or write on that particular block. As a result, random read/write operations are taxing.

A device that writes 10KB of random data 5000 times in one second is much faster than a device that reads sequential data of 10KB 10,000 times in one second. However, it goes without saying that you will buy the device based on your workload. If your workload requires sequential reads, it will be a bad investment in terms of ROI to buy a device on the basis of its random write performance.

5. **Queue depth:** In a Utopian world, there would just be one thread doing read and write operations. However, in the real world, the IO world consists of hundreds or even thousands of threads performing IOs on multiple disks. These disks are connected to a controller which, in turn, could be connected to another controller. At its basic unit, an IO is just a SCSI (or relevant protocol) command sent to a SCSI device to perform an operation on a SCSI disk. Every IO request is an entry in the queue of the device. Queue depth is the maximum number of IO requests that can be executed at one time. The longer the queue depth, the larger the number of IO requests that can be submitted. However, if the device is not able to process those requests faster, we might end up in a QUEUE_FULL state, which results in increased latency. Queue depth is a tunable parameter from the OS controlling the disk controller. If the latency of a device is very high, it makes sense to examine its queue depth.
6. **Throughput:** I have deliberately kept this metric for the end as most people are familiar with it and evaluate a storage device based on throughput. To put it simply, throughput is the product of IOPS and IO size. So if a device reads and writes 4KB of data 10,000 times in one second, its throughput will be 40Mbps.

Then there are partial IOs, synchronous or asynchronous IOs, direct or cached IOs and a few other variants. However, they shift the gear towards application-specific workloads. A deep understanding of these performance indicators can help users make the right choices in storage devices, and create effective strategies to configure and tune the system such that there are no bottlenecks in an IO path, and applications give optimal performance and work seamlessly. Learning about configurations and tuning strategies could be a part of our next article on performance. END

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